Hashtag AI at HalluScoring-2026: Distilling Knowledge-Graph Retrieval into Gemma 4 for Arabic Hallucination Detection*

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Abstract

We describe our submission to HalluScoring 2026 for Arabic hallucination detection in general-culture and Islamic-knowledge question answering. Our system adapts the instruction-tuned 12B-parameter Gemma 4 model with QLoRA. Its central idea is to use the Hashtag AI knowledge-graph retrieval system only during offline data construction: we query facts related to development-set failures, create contrastive training examples, and distill them into the model parameters. The submitted model is closed-book and uses no retrieval, search, or external knowledge at evaluation time. On organizer-provided test annotations, it obtains 73.59 macro-F_1 on the 400-example Islamic set and 77.56 on the 1,600-example mixed-domain set; answer-option accuracies are 75.25% and 81.13%. Analysis shows strong science performance but lower recall for subtle hallucinations in history, geography, and Islamic knowledge.

1 Introduction

Hallucination detection is especially consequential for Arabic question answering because a fluent response can conceal a factual error, and errors in Islamic content may concern sensitive religious claims. HalluScoring 2026 (Alansari et al., 2026) evaluates systems on general-culture and Islamic-knowledge data using Arabic question–answer pairs. We participate in both Tasks 2.1 and 2.2.

Our approach is deliberately compact: we adapt a sub-13B instruction model rather than relying on a larger proprietary model at inference. We fine-tune Gemma 4 12B IT with quantized low-rank adaptation (QLoRA; Dettmers et al. 2023), constrain the output to a small JSON schema, and use deterministic decoding. Knowledge-graph retrieval serves as an offline teacher for difficult facts, not as an evaluation-time component.

Knowledge graphs represent entities and their relations in a structured form that supports querying and reasoning (Hogan et al., 2021). Retrieval-augmented generation (RAG) complements a model's parametric memory with retrieved non-parametric evidence (Lewis et al., 2020); more recent work extends this idea to question answering over textual graphs (He et al., 2024). Unlike these inference-time retrieval systems, our submitted detector performs no retrieval at evaluation. We use knowledge-graph evidence only to construct additional supervision, then distill that supervision into the model weights.

Our main contributions are: (i) a rule-compliant method for distilling knowledge-graph evidence into a closed-book detector; (ii) a reproducible parameter-efficient Gemma 4 system; and (iii) a domain-level analysis of correct and failed cases.

2 Task and Data

Each instance contains an Arabic question, a generated answer, and a binary label: `hallucination` or `no_hallucination`. Task 2 instances additionally provide six candidate answers (A–F), and the submission format requires both a predicted label and a selected option. The shared-task benchmark is HalluTruthQA (Bouchekif et al., 2026); the Track 2 resource is its HalluTruthQA-4K corpus (Bekhouche et al., 2026), comprising 4,000 expert-curated instances across Islamic knowledge, history, science, and geography. The organizers supplied 400 Islamic-knowledge test examples for Subtask 2.1 and 1,600 examples for Subtask 2.2; the latter comprises four 400-instance groups in the delivered file. To prevent leakage, inference files retained only the ID, question, generated answer, and options; all labels, reference answers, and annotations were removed before prediction and used

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only for post-hoc local evaluation.

For supervised adaptation we use the 4,705 labeled instances distributed in the official Track 1 starter kit, which is associated with the HalluScore Arabic hallucination benchmark (Alansari and Luqman, 2026). A fixed seeded split produces 4,215 training and 490 validation examples (89.6/10.4%). The training portion contains 2,214 hallucination and 2,001 non-hallucination examples; the validation portion contains 259 and 231, respectively. We keep the official development sets (1,300 examples for Task 1.1 and 200 for Task 1.2) disjoint from training and use them only for model comparison.

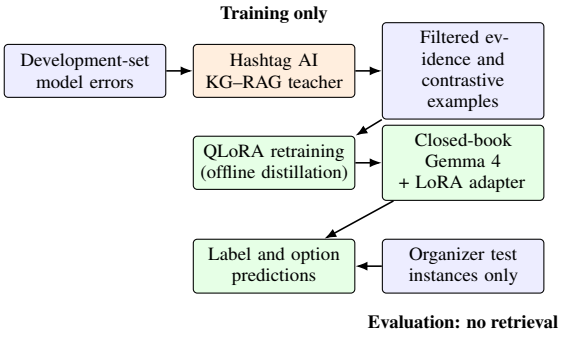


Figure 1: Knowledge-graph evidence is used only to construct and distill offline supervision. The submitted model receives no KG, retrieved evidence, search results, or external API access at evaluation time.

3 System Description

3.1 Detection Model

The backbone is `google/gemma-4-12B-it`, selected to satisfy the task's model-size constraint. We express each example as a chat instruction containing the question and generated answer. The system instruction requests exactly one JSON object and no explanation. Its single `predicted_label` field contains either `hallucination` or `no_hallucination`. Only assistant tokens contribute to the causal language-modeling loss.

We use LoRA (Hu et al., 2022) on the attention projections (`q`, `k`, `v`, and `o`) and MLP projections (`gate`, `up`, and `down`). The rank is 16, scaling factor 32, and dropout 0.05. The frozen backbone is loaded in 4-bit NF4 with double quantization and bfloat16 computation. We train for three epochs with a maximum length of 2,048 tokens, batch size 1, 16-step gradient accumulation, learning rate 2×10^{-4}, cosine scheduling, 3% warm-up, and paged 8-bit AdamW. This setup follows the memory-efficient QLoRA recipe while limiting learned parameters to the adapter.

3.2 Offline Knowledge-Graph Distillation

The shared-task rules permit external data for fine-tuning but prohibit RAG, search, knowledge bases, and external information during evaluation. We therefore separate knowledge acquisition from inference. This design is motivated by RAG's use of retrieved evidence (Lewis et al., 2020) and by graph-based retrieval for knowledge-intensive question answering (He et al., 2024), but moves retrieval entirely to offline supervision construction. After evaluating the first QLoRA model, we collect questions from its remaining errors, focusing on history and geography. During data preparation only, we submit these questions to the knowledge-graph-based reasoning and RAG system developed by Hashtag AI (Hashtag AI, 2026). Retrieved answers are canonicalized into short references, manually filtered, and converted into both faithful answers and hard hallucinations through controlled perturbations of dates, quantities, entities, locations, and unsupported details.

Figure 1 summarizes the separation between the offline teacher pipeline and evaluation-time inference.

The final augmentation is grounded in 9 answered hard questions and contains 100 examples (55 hallucinated and 45 non-hallucinated). We repeat this focused set ten times and mix it with 4,215 official training examples, yielding 5,215 training rows; the 490-example official validation split remains unchanged. This resembles a student reviewing externally explained mistakes before a closed-book examination: the knowledge graph teaches the model during training, but neither retrieved passages nor the KG system are available to the submitted model.

At evaluation time the artifact consists only of the frozen Gemma 4 backbone and learned LoRA weights. Prompts contain only organizer-permitted instance fields. No API call, retrieval index, external search, or knowledge-base lookup occurs, satisfying the competition's closed-book constraint.

3.3 Inference and Option Selection

Inference is greedy and deterministic. A parser first reads the requested JSON key and normalizes common surface variants. If parsing fails, the detector conservatively emits `hallucination`. For Task 2,

Model	Training data	Evaluation split (n)	Acc.	Macro-F_1	Option Acc.
Gemma 4 12B	None (zero-shot)	General dev (1,300)	91.31	91.20	–
Gemma 3 12B SFT	Official (4,215)	General dev (1,300)	88.69	88.43	–
Gemma 4 12B QLoRA	Official (4,215)	General dev (1,300)	**95.62**	**95.53**	–
Gemma 4 KG-QLoRA	Official + KG (5,215 rows)	General dev (1,300)	95.15	95.06	–
Gemma 4 QLoRA	Official (4,215)	Islamic dev (200)	97.00	96.76	–
Gemma 4 KG-QLoRA	Official + KG (5,215 rows)	Islamic dev (200)	**97.50**	**97.31**	–
KG-QLoRA final	Official + KG (5,215 rows)	Test 2.1 (400)	76.25	73.59	75.25
KG-QLoRA final	Official + KG (5,215 rows)	Test 2.2 (1,600)	79.31	77.56	81.13

Table 1: Training data and evaluation results (percent). *Official* denotes the 4,215-example training split; *Official + KG* adds 1,000 repeated rows drawn from 100 manually filtered KG-derived examples. Accuracy and macro-F_1 evaluate hallucination detection; option accuracy evaluates A–F answer selection and therefore applies only to Task 2.

a second prompt receives the question, generated answer, candidate options, and predicted detection label, and emits a single option letter. A malformed option response falls back to the candidate with the highest normalized character-sequence similarity to the generated answer. This fallback is intended only to guarantee a valid submission schema.

The released notebook loads the adapter, runs both stages, and creates the prediction, metadata, and submission files. Code, prompts, and configurations are available at `https://github.com/Anugithub2019/Koran`.

4 Results

We report accuracy and macro-F_1, with macro-F_1 giving equal weight to both labels. Table 1 summarizes completed runs. Plain QLoRA strongly improves over the prompt-only backbone. KG-guided retraining produces a mixed result: relative to plain QLoRA it improves Task 1.2 accuracy from 97.00% to 97.50%, but decreases Task 1.1 from 95.62% to 95.15%. We use the KG-guided model for both Task 2 test submissions. Local test evaluation uses organizer-provided annotations; official Codabench scores were not yet available at writing time.

On Test 2.1, the detector produces 216 true negatives, 24 false positives, 71 false negatives, and 89 true positives. On Test 2.2, the corresponding counts are 858, 59, 272, and 411. Hallucination precision is high (78.76% and 87.45%), but recall is only 55.63% and 60.18%, respectively. Thus, the main error is accepting a plausible but hallucinated answer.

5 Results Analysis

Correct cases. The detector is strongest when the generated answer directly conflicts with a stable fact, such as a wrong date, named entity, location, or scientific definition. Science is the clearest success case on Test 2.2: label accuracy is 93.75%, macro-F_1 is 93.64, option accuracy is 93.00%, and hallucination recall is 90.45%. Across the complete test sets, non-hallucination recall is 90.00% (Task 2.1) and 93.57% (Task 2.2), showing that faithful answers and paraphrases are usually accepted.

Failure cases. Errors are strongly domain-dependent. Geography reaches 71.50% accuracy and only 53.62% hallucination recall. The 55-instance history subset identified by the `historic2` prefix is hardest: accuracy is 50.91%, macro-F_1 is 41.56, and only 3 of 30 hallucinations are detected. The remaining 345 history cases fare better (79.71% accuracy), but hallucination recall remains 43.52%. Islamic knowledge obtains 76.25% accuracy and 55.63% hallucination recall.

Inspection reveals four recurring patterns. First, topically plausible answers naming a neighboring entity, date, location, or religious source are often accepted. Second, the model accepts broadly defensible alternatives when the reference expects one specific formulation. Third, underspecified questions encourage reasonable caveats that nevertheless disagree with the short reference. Fourth, partially correct answers combine a true core claim with an unsupported detail, making the hallucination subtle. Option selection is less affected than binary detection on geography and the small history subset, suggesting that the model often identifies the best reference answer even when it fails to label the generated answer as hallucinated. Output-format fallbacks were needed for 13/400 labels and 14/400 options in Task 2.1, and 55/1,600 labels and 42/1,600 options in Task 2.2.

The precision–recall imbalance clarifies the aggregate scores. Relative to non-hallucinated answers, hallucinated answers are more heterogeneous: an error may alter one entity, append one unsupported clause, or exploit ambiguity in the question. The detector's high hallucination precision but moderate recall therefore indicates conservative positive predictions rather than indiscriminate over-flagging. The gap between binary-label and option accuracy further shows that ranking a supplied reference among six candidates is easier than deciding whether a free-form answer contains any unsupported claim. Future augmentation should consequently emphasize minimally edited, partially correct negatives and ambiguous cases rather than only obvious factual substitutions.

6 Limitations and Conclusion

Our test scores are local calculations from the annotations supplied with the corpus and must be replaced or confirmed by official Codabench results. The small `historic2` subset makes its topic estimate uncertain, and its distinct ID prefix may indicate a distribution shift. Parser fallbacks can also conflate generation-format errors with classification errors. The model is closed-book and may reproduce knowledge gaps or biases from its backbone. Finally, religious claims warrant expert verification; a binary classifier should not be treated as an authority on Islamic knowledge.

We presented a compact Gemma 4 QLoRA system that converts knowledge-graph retrieval into offline supervision and remains fully closed-book at evaluation time. The KG-guided augmentation improves the smaller Islamic development split but does not uniformly improve every domain, motivating broader and more diverse knowledge distillation. Test analysis identifies semantic near-misses, underspecification, partial correctness, and annotation granularity as the main remaining challenges.

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A Related Work

Arabic hallucination benchmarks. HalluScoring builds on recent efforts to evaluate factuality in Arabic question answering. HalluScore studies Arabic hallucination detection across general and Islamic knowledge (Alansari and Luqman, 2026), while HalluTruthQA introduces fine-grained detection, localization, and explanation annotations (Bouchekif et al., 2026). HalluTruthQA-4K further documents the corpus construction and expert annotation process used for truth verification (Bekhouche et al., 2026). Our work differs from benchmark construction: we focus on a constrained shared-task system and analyze how a compact, closed-book detector behaves across domains.

Parameter-efficient adaptation. LoRA learns low-rank updates while keeping the pretrained backbone frozen (Hu et al., 2022); QLoRA combines this idea with quantized backbone weights to reduce fine-tuning memory (Dettmers et al., 2023).

We apply QLoRA to an instruction-tuned Gemma 4 model and supervise a constrained JSON classification output. This makes the final artifact reproducible and deployable without a retrieval service.

Retrieval and knowledge distillation. RAG supplements parametric knowledge with retrieved evidence at generation time (Lewis et al., 2020), and graph-based retrievers extend retrieval to structured or textual graphs (Hogan et al., 2021; He et al., 2024). Our use of retrieval is deliberately different: a KG–RAG system acts only as an offline teacher for constructing additional training examples. At evaluation time the student receives only the organizer-provided fields. The contribution is therefore best viewed as retrieval-to-parameter distillation under a closed-book evaluation constraint, rather than an inference-time RAG system.

B Additional System and Data Details

Table 2 makes the supervision mixture explicit. The KG-derived set contains 100 unique examples grounded in nine answered hard questions. We repeat this focused set ten times so that rare, difficult patterns have non-negligible weight during the second training stage. Repetition changes the number of training rows but not the number of unique KG-derived examples.

Source	Unique	Rows used	Hall./Non-hall.
Official training split	4,215	4,215	2,214/2,001
KG-derived augmentation	100	1,000	55/45 unique
Final training mixture	4,315	5,215	–
Official validation split	490	490	259/231

Table 2: Supervision used by the final KG-QLoRA model. *Hall.* abbreviates hallucination. The validation split is never included in training.

Fine-tuning objective. For a target linear layer ℓ, let $W_\ell \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ be its frozen pretrained weight. QLoRA stores the backbone in 4-bit form; $Q_4(W_\ell)$ denotes the corresponding dequantized weight used in the forward pass. It learns only a rank-r update,

$$W_\ell^{\text{eff}} = Q_4(W_\ell) + \frac{\alpha}{r} B_\ell A_\ell.$$

The factors have dimensions $A_\ell \in \mathbb{R}^{r \times d_{\text{in}}}$ and $B_\ell \in \mathbb{R}^{d_{\text{out}} \times r}$. Here $r = 16$ and $\alpha = 32$. If y_{it} is the target token at position t in example i, the trainable adapter parameters $\theta = \{A_\ell, B_\ell\}_\ell$ minimize masked causal cross-entropy,

$$\mathcal{L}(\theta) = -\sum_i \sum_t m_{it} \log p_\theta(y_{it} \mid x_i, y_{i,<t}),$$

where $m_{it} = 1$ only for assistant-response tokens. Thus, the question, generated answer, and instruction provide context, while loss is applied only to the target JSON output. The quantized Gemma backbone remains frozen throughout training.

Knowledge graph and its role. A knowledge graph can be represented as a directed labeled multigraph $G = (V, E)$, where nodes V denote entities or concepts and an edge $(u, r, v) \in E$ records a relation r, for example a person–place, event–date, or work–author fact. For a development error with question q, the offline teacher retrieves a relevant subgraph $G_q \subseteq G$ and verbalizes its relations as evidence z_q. We manually filter z_q, form a faithful answer, and construct hard hallucinations by perturbing a relation endpoint, date, quantity, location, or unsupported clause.

This structure helps in two ways. First, explicit relations disambiguate neighboring entities that a language model may otherwise conflate. Second, the same verified fact can generate controlled contrastive pairs that differ in only one factual relation, directly targeting the detector's failure modes. Training transfers this supervision into the LoRA parameters; neither G_q nor z_q is available at evaluation. The mixed development result in Table 1 also limits the claim: the focused graph-derived set improves Islamic development performance but is too small and narrow to guarantee gains in every domain.

The detector and option selector are run sequentially. The first prompt maps the question–answer pair to one of two labels; the second maps the same instance and six supplied options to a letter. Greedy decoding removes sampling variance. Parsing first targets the requested JSON key, then normalizes surface variants; only malformed generations invoke the conservative label or similarity-based option fallback described in Section 3.3.

C Additional Qualitative Analysis

Table 3 gives stylized examples of the recurring behaviors observed during manual inspection. They abstract the semantic pattern rather than reproduce corpus items verbatim, so they should not be interpreted as additional scored instances.

The domain results suggest that augmentation coverage matters as much as model capacity. Science questions often contain crisp definitional or numerical conflicts, whereas history, geography, and Islamic knowledge more often require resolv-

Pattern	Typical outcome	Diagnostic interpretation
Neighboring entity	False negative	A related ruler, scholar, city, or source remains topically plausible.
Wrong date/quantity	Mixed	Salient contradictions are detected; small numeric changes are less reliable.
Partial correctness	False negative	One unsupported clause is masked by an otherwise true answer.
Reference granularity	False positive/disputed	A defensible answer may be broader than the expected formulation.
Faithful paraphrase	Correct negative	High non-hallucination recall shows tolerance of rewording.
Option/label disagreement	Correct option, wrong label	Candidate ranking is easier than free-form factuality detection.

Table 3: Representative qualitative patterns abstracted from manual error inspection.

ing closely related entities, chains of transmission, locations, or competing formulations. The especially low recall on the small `historic2` subset should be treated cautiously because its sample size is only 55 and its identifier suggests a possible distribution shift. Nevertheless, the repeated failure on partially correct answers points to a concrete next step: generate minimal-pair examples in which exactly one clause, entity, date, or attribution is perturbed while the remainder of the answer is preserved.